Final Report Structure

Page 1: Title & Cover Page

- **Student Details: Name:**

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Submission Deadline:

15-11-2026

- **Project Title:** Sales Analytics & Business Intelligence System

- **Deliverables Included:** Excel, SQL Backup, SQL Scripts, Power BI (.pbix), Screenshots, Business Summary

Page 2:Introduction

- **Business Problem:** The enterprise requires a centralized, robust sales analytics and business intelligence system to evaluate commercial transactions, monitor financial performance, track order returns, and assess logistics operations. Without structured insights, management cannot effectively identify operational inefficiencies or profit leakage.

- **Project Objectives:**

1. Inspect, clean, and standardize raw transactional files (`Orders`, `Persons`, `Returns`) using Microsoft Excel.
2. Build a normalized relational database in SQL Server (`Sales_Analytics`) to maintain referential integrity and generate reporting views.
3. Develop an interactive star-schema Power BI dashboard equipped with core financial measures, operational metrics, and advanced What-If scenario modeling.

- **Dataset & Tool Overview:**

  - *Source Files:* SuperStore – Orders.xlsx, SuperStore – Persons.xlsx, and SuperStore – Returns.xlsx.
  - *Technology Stack:* Microsoft Excel (Phase 1), SQL Server Management Studio / T-SQL (Phase 2), and Power BI Desktop (Phase 3 & 4).

Page 3 — Excel Preparation (Phase 1)

- **Data Quality Issues Identified:**
  - Presence of duplicate order line items within transactions that required careful grain verification.
  - Inconsistent regional labels, formatting anomalies in date fields, and missing values in critical business columns.
- **Cleaning Methodology:**

    - Standardized spelling and casing across `Region`, `Category`, and `Segment` attributes.
    - Reconciled row counts before and after deduplication to maintain strict accounting accuracy.
    - Engineered a derived shipping delay attribute (`Shipping_Delay_Days = Ship Date - Order Date`) while filtering out invalid date entries.

- **Validation Summary:**
    - The cleaned workbook was successfully saved as `Sales_Cleaned_Data.xlsx` with distinct, standardized sheets ready for database ingestion.

Page 4 — Database Engineering (Phase 2)

- **Database Architecture:** Created a centralized SQL Server database named `Sales_Analytics`.
- **Table Design & Grain:**
    - `Orders`: Fact table structured at the individual line-item grain.
    - `Persons`: Lookup dimension mapping unique regions to assigned personnel.
    - `Returns`: Order-level tracking table mapped via `Order_ID`.
- **Relational Integrity & Views:**
    - Implemented primary and foreign key constraints to enforce referential integrity and prevent data anomalies.
    - Authored a consolidated reporting view named `vw_Sales_Final` incorporating calendar attributes, geography, category, financial measures, and shipping status to streamline Power BI connectivity.

Page 5 — Dashboard Development (Phase 3 & 4)

- **Data Model & Star Schema:** Configured a robust star schema in Power BI Desktop with `Orders` acting as the central fact table linked to optimized descriptive dimensions.
- **Core DAX Measures:**
    - *Total Sales & Total Profit:* Evaluated overall financial health.
    - *Total Orders:* Computed true business order counts rather than raw line-item occurrences.
    - *Profit Margin % & Return Rate %:* Tracked profitability ratios and order-level return percentages.
- **Advanced Decision-Support Measures (Phase 4):**
    - `Profit Gain from Discount Cut` ($231.68K) vs. `Profit Drop from 5% Extra Discount` ($579.20K).
    - `Recovered Return Revenue (10% Cut)` ($67.79K).
    - `Required Sales Growth for 10% Target` ($1.16M).
- **Visual Layout & Interactivity:** Designed a dedicated "Advanced Scenarios & Decision-Support" page featuring 4 primary KPI cards, comparative category/region bar charts, a shipping delay risk scatter plot with category legends, and synchronized global slicers (`Region`, `Category`, `Year`, `Month`).

Page 6 — Business Findings (8 Core Questions)

1. **Highest Sales Region:** Empirical analysis confirms that the **West** and **East** regions generate the highest overall sales volume across the enterprise.
2. **Highest Profit Category:** The **Technology** category produces the highest cumulative profit margin, driven by high-value product lines.
3. **Loss-Making Products:** Specific items such as the *Cubify CubeX 3D Printer* and select regional furnishings contribute disproportionate financial losses due to deep discounting.
4. **Monthly Sales Trend:** Sales exhibit strong seasonality with upward momentum peaking towards the end of Q3 and Q4.
5. **Return Rate Variation:** Return rates vary significantly by region, with **North Asia (13.93%)** and **North (10.20%)** recording the highest return percentages.
6. **Highest Profit Customer Segment:** The **Consumer** segment delivers the largest share of total profit, followed closely by Corporate accounts.
7. **Shipping Delay vs. Profit:** Standard Class shipments experience the longest delivery delays (averaging ~5 days), creating localized customer retention risks.
8. **Strategic Overview:** Baseline performance metrics combined with Phase 4 simulations indicate immediate opportunities for margin optimization through targeted operational controls.

Page 7 — Scenarios & Recommendations (Phase 4)

- **Assumptions & Methodologies:** Scenario evaluations are treated as tactical estimates derived from historical baseline sales ($11.58M) under the assumption of stable demand elasticity.

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- **Prioritized Action Plan:**
  - **High Impact / Low Effort:** Implement strict quality control measures in **North Asia** and **North** regions to achieve a 10% reduction in returns, recovering approximately $67.79K in revenue.

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  - **Margin Protection:** Apply strict regional discount limits on loss-making items to secure a potential profit gain of $231.68K.

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  - **Logistics Optimization:** Streamline logistics for Standard Class shipments to mitigate delivery delays and protect customer satisfaction.

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- **Limitations & Future Scope:** Future enhancements should incorporate real-time courier cost tracking, detailed customer churn analysis, and empirical price-elasticity coefficients to refine predictive accuracy.